Housing First or Treatment First?

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Response to the Associate Editor

Comments A1 to A20, responses in red

Thank you for a letter that was unusually specific about what a successful revision would have to contain. The three headings you set out, careful modelling of the mechanisms, careful modelling of the policies, and careful quantification, are the organising principle of the revision.

One note before the comments. The paper has a new title: it was submitted as *Housing First or Treatment First?* and is now *Housing First, Children Later*. The reason is the substance of the revision rather than a relabelling, and it will be clearest at A6, since the mechanism the title names is the one you and Referee 2 asked for.

Careful modelling of the key mechanisms

A1 A stylised model to convey the mechanisms

> *"In order to convey the different mechanisms of the paper, it would be helpful to provide a section with a simplified version of the model as suggested by both R1 and R3. The sheer number of moving parts are currently obfuscating the economics of the paper. . . . Derive basic model predictions on the relationships between the key model outcomes (income, unemployment, housing, drug use) and show these align with a few stylized facts."*

Done, and with one change of substance relative to what the previous appendix did. The old appendix was static. Referee 1 comment 1e makes the point that a static object cannot establish claims about a dynamic mechanism, and he is right: the paper's mechanism is that drug use lowers the *continuation* value, which a static model cannot represent.

The new Section 3 is therefore a stripped down *dynamic* model: two periods, one asset, a binary use choice, a binary housing outcome, and a survival probability that depends on both. It is small enough to solve by hand, and it is also calibrated, to the same United States moments as the quantitative model, so the results below are numbers rather than signs: six of its parameters are set inside it, against drug participation, the adult substance use disorder rate, the homelessness rate, the value of a statistical life, the housing expenditure share, and the dispersion of earnings. The last of those is calibrated to the same moment in both models, so the two carry the same income dispersion by construction rather than by coincidence. It delivers the three results the paper needs:

1. Use falls in permanent income whenever the penalty for use is a mortality risk, and rises in permanent income when the penalty is proportional to income. This is the paper's identification argument, and it makes clear why the sign of the income gradient in the data is informative about which channel dominates.
2. Guaranteed housing raises the continuation value and therefore lowers use; unconditional transfers relax the constraint and therefore raise it. The two forces have opposite signs, which is why the net effect is a quantitative question.

3. The cross partial of welfare in generosity and conditionality is positive. This is the paper's central claim stated as a comparative static: conditionality is worth more when the benefit being conditioned on is guaranteed than when it is rationed. It also explains why the previous draft's Experiment 1, which varied conditionality under rationing, produced the least attractive of the four regimes.

The stylised facts are presented alongside, as you and Referee 3 both ask.

A2 Is modelling unemployment insurance expiration important?

"Is it important to model the expiration of unemployment insurance?"

[Conceded.] No. The benefit duration state is cut. Nothing in the calibration identified it and no policy conclusion moved with it.

A3 The role of labour supply

"What is the role of labor supply in this framework?"

Kept, but coarsened, and now visible in the results.

I flag a genuine disagreement between you and Referee 3 here. His minor comment 1 says he is "surprised that there is not a clearer role for the labor supply elasticity in welfare calculations, as it should matter quite substantially for the welfare costs of balancing the budget with extra labor taxes". Since every counterfactual in the paper is financed by the labour income tax, I think he is right that it belongs, and that the previous draft's failure to show its role is what made it look ornamental.

The fix is to show it rather than to argue about it. The welfare decomposition now separates the mechanical transfer from the deadweight loss of the tax change, by re solving the counterfactual with the hours choice frozen at its benchmark schedule. Freezing the choice set preserves the Bellman identity, which the previous approach of putting a large penalty on the disutility of labour did not: that drove hours to a corner rather than holding them fixed. The deadweight column is now a clean number, and a reader can see immediately whether labour supply is doing any work.

A4 Why homeownership rather than homeless against housed?

"Why do we need the role of homeownership, and not simply the difference between homelessness and houses?"

[Conceded.] Cut. The housing state is now homeless plus three rental sizes. This also resolves Referee 2's detail comment D1: the previous grid had seven points of which three were calibrated and the rest were listed in a footnote. Every point of the new grid is pinned by a moment, and the moments are the size distribution of occupied units from the American Community Survey rather than the budget consequences of the grid.

A5 Why full general equilibrium?

"Why does the policy analysis require a full general equilibrium model? Many long-run partial equilibrium outcomes appear nearly identical to the general equilibrium ones. Do account for the

changes in the government, and ideally for the adjustment in drug prices (See R1 Comment 1c) not necessarily in terms of housing prices and rents."

Done exactly as specified. House prices and rents are now fixed. The government budget still clears through the labour income tax, and the drug price now clears a supply curve. Two prices out, one in.

One consequence is worth flagging because it changes how the paper reads. With the drug price endogenous, a policy that reduces the number of users lowers demand and therefore lowers the price, which partially offsets the reduction. The size of that offset depends on the elasticity of drug supply, which is not a parameter with an obvious value, so I have gone to some length to discipline it. There is no direct estimate for an illicit drug. Cannabis is the closest market to one, because it remains a Schedule I substance federally, so even a state licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. Mace, Patel and Seegert (2020) estimate a price elasticity of supply of 1.46 on Washington State administrative data, and that is the number the model uses. Cigarettes and gasoline share none of those frictions, so I do not bracket with them. I explain in the text why the frequently quoted RAND figure of minus 3.6 is not the right object here: a negative supply elasticity comes from enforcement swamping, which Caulkins and Reuter note "pertains to total drug volume ... not just or primarily drug specific market volume", so it operates on the aggregate market rather than at my margin.

A6 Intergenerational transmission discussed without intergenerational linkages

"Why is there a discussion on intergenerational transmission in Section 4.1.3, without intergenerational linkages?"

[Conceded.] You were right that the discussion was decorative. Referee 2's comment C3 asks for the opposite fix, an actual linkage, on the ground that parental anxiety about children's outcomes would amplify the welfare gains from reducing addiction.

[New result.] I made the linkage real rather than deleting the discussion, and it has since become the organising idea of the paper, which is why the paper now has a different title. Your comment and Referee 2's C3 are the origin of it: you objected that the transmission discussion had nothing in the model behind it, Referee 2 asked for exactly the linkage that would give it something, and building it turned out to decide the paper's answer rather than decorate it. A household that acquires the disorder, or that ends up on the street, passes a worse productivity draw to the child who replaces it and counts only a fraction ω of that loss. The remainder is an externality. It is the one that decides the comparison: Housing First weakly dominates Treatment First for every household taken one at a time, at the same generosity and the same cost, and the ranking reverses once the children of the households it fails to deter are counted.

Three things make it load bearing rather than decorative.

It costs no new state. The continuation splits into the household's own next period, weighted by survival, and the child who replaces it, weighted by the chance of dying and by ω. The child term is a difference, the value of a child born to this household against one born to a clean and housed parent, so it is zero in the reference state and negative otherwise, and only the loss the parent's own state imposes enters the decision.

Each link is measured. Parental substance abuse is recorded at more than a third of removals into out of home care; children of a parent with a disorder are two to four times as likely to acquire one;

a parental shock reaches the child's adult earnings, which Nilsson (2017) puts at about twenty four per cent for prenatal exposure to an alcohol availability experiment; and housing is one of the environments doing the transmitting, which Chetty et al. (2016) put at thirty one per cent for a move to a lower poverty neighbourhood before age thirteen.

And ω is calibrated rather than chosen. Every household in the model has a child, so there is no parent against non parent comparison in the benchmark and the moment has to be a counterfactual. In NSDUH 2024, past month use among parents is 0.845 of use among non parents, reweighted to the non parents' joint distribution of age, income, sex and marital status, and ω is set so that removing the child from the household's problem raises use by the same 18.3 per cent.

I report results with ω set to zero as well, so a reader can see exactly how much rests on it. Referee 2's expectation was that the channel would amplify the gains from reducing addiction. In this model it does more than amplify: it is what reverses a ranking that is otherwise unambiguous.

A7 Lump sum transfers and type dependent bequests

> *"Incidentally, how are the lump-sum transfers determined as a function of types? And why is it important that the bequests are type-dependent?"*

[Conceded.] It was not important. Both are now uniform. There was no argument for the type dependence and no moment identified it.

A8 Preferences: keep the weights invariant and vary b instead

> *"A parsimonious way that came to my mind is to keep the Cobb-Douglas weights invariant to addiction but let the marginal utility vary by addiction by changing the parameter b decrease by addiction status."*

I have gone further than this, and I want to explain why, because your suggestion is the natural one and it does not quite work.

Referee 2 comment C5 and Referee 3 comment M2a identify the same problem from different sides: making the utility weight depend on substance use disorder bakes in an income contingent gain or loss from acquiring the disorder, and it makes reducing addiction and expanding housing mechanical welfare complements even with no incentive effect at all. That is a serious charge, because the complementarity between generosity and conditionality is the paper's headline.

Varying b alone, holding the weights fixed, removes the mechanical complementarity but cannot reproduce the target it was doing the work for. The share of income spent on drugs is 44 per cent among people with a disorder and 11 per cent among experimenters. With a common weight, b shifts the *level* of drug spending, but the ratio of the two spending shares is governed by the curvature of the subsistence term, and moving b far enough to open a fourfold gap drives experimenters to a corner.

The revision therefore fixes preferences *fully* across disorder status: one weight, one b. The 44 against 11 gap then has to emerge endogenously, from forward looking restraint. A household without a disorder limits its use because use raises the probability of acquiring one; a household that already has one faces no such margin. That is a stronger result than the one it replaces, because the spending gap becomes a prediction rather than an assumption, and the welfare complementarity becomes an incentive effect rather than an artefact of the utility function.

[Built since the first draft of this letter.] There is a deeper reason than the calibration one, and it is what settled the choice. Any specification that makes a household *more* susceptible to drugs by moving its preferences also makes that household *better off*, mechanically, because it is now getting more utility from something it was going to consume anyway. Flow utility here is Cobb-Douglas in $b + v$, so the marginal value of the drug is proportional to $1/(b + v)$: a smaller b raises the marginal value of using and raises desired intake, and it raises the level of utility along with it. That sign is the wrong way round against everything known about substance use disorders, which are not experienced as a windfall by the people who have them, and it would put the welfare accounting on exactly the wrong footing: the policy that prevents a disorder would be taking something valuable away.

So the model varies *income* rather than utility. Preferences are one weight and one b, identical in every state and after every choice, in the code as well as in the text: the function that returns the shifter takes the disorder and the dwelling as arguments and returns the same number for all of them, and the two Cobb-Douglas weights are set equal by construction with the switches that would separate them held at zero. What the disorder changes is the budget and the hazards it faces, through lost earnings, lost employment, higher mortality and harder recovery. Those are measured objects, and every one of them makes the household worse off, which is the direction the evidence points.

The same argument governs what the child inherits, and it is worth saying because it is the paper's central mechanism. A child of a user inherits a worse productivity draw, not a stronger taste. Had it inherited the taste, through a lower b, its lifetime value would have gone *up*, so the specification would say that a child of a parent with a disorder is better off for it. That is the opposite of what these children report.

A9 Loss of consortium is ad hoc; present welfare and mortality separately

> *"The consortium adjustment modeling also seems very ad-hoc while at the same time mattering a lot of the welfare ranking as noted also by the Referees. A possible way to address this would be to instead present the policy consequences along two different dimensions, the welfare consequences of equation (30) and the mortality consequences."*

[Conceded.] The parameter is deleted. Referee 1's version of this objection, that a truly conservative value for a number plucked from the air is zero and that zero reverses the ranking, is unanswerable and I do not try to answer it.

I do both things you suggest. Welfare and mortality are reported as two dimensions. And, following Referee 2's comment C2, the externality that the loss of consortium adjustment was gesturing at is now inside period utility as a disutility from the size of the homeless population, so households internalise it. This means the welfare ranking is produced by the model rather than added to it afterwards, which was Referee 1's actual complaint.

[Built since the first draft of this letter.] The two dimensions are now reported separately throughout, and the paper's conclusion has moved with them. The defensible result after this revision is about *generosity*: expanding coverage dominates, and conditionality is second order and uncertain. That is consistent with the quasi experimental evidence you point to, which says the mortality benefit runs through housing and that conditioning on treatment adds nothing detectable.

Modelling of the policies

A10 Treatment First actively treats; the model has no treatment

> *"Modeling 'Treatment First' as a voucher program that depends on sobriety seems to omit key characteristics. As R1 Comment 1 discusses, the program actively aims to cure substance use disorder before housing individuals. The policy itself has a direct role in impacting the transition out of addiction."*

[Conceded.] This and Referee 1's comment 1b are the same point and it is the most damaging one in the reports: a policy named Treatment First contained no treatment, and recovery was a coin flip that no policy could touch.

Households with a substance use disorder now choose whether to take up treatment. Take up raises the recovery hazard. The cost is in *utility* rather than in resources, deliberately: a resource cost comes out of the same budget that buys food and rent, so the households with the strongest case for treatment, the poorest, would be exactly the ones who could not afford it, which is a modelling artefact rather than a feature of the world.

The utility cost is identified by treatment take up among people with a substance use disorder, roughly a quarter in NSDUH. This was previously a free parameter with no source at all, which is uncomfortable given that the entire Treatment First result now runs through it.

Referee 1 predicts this addition will *strengthen* Treatment First. I report what the model says rather than what I hoped.

A11 The policy is unlikely to be perfectly enforceable

> *"Moreover, the policy is unlikely perfectly enforceable. R1 Comment 4 suggests additional features of 'Treatment First' that are of first-order importance for the policy evaluation."*

Done, and put in the budget constraint rather than in a footnote. Each voucher holder faces a per period probability of being tested; each test costs the government money; detection can only happen if a test happened; and clean households can be flagged in error. The previous specification is the corner where the testing probability equals one.

Referee 1's comment 4c makes the sharper version of this: the previous draft dismissed a one to two billion dollar annual testing cost in a footnote, and he argues that the estimate is both probably far too low *and* too large to ignore even at its stated value. Both halves are right. The cost now competes with everything else the government pays for, which is where it should compete.

Quantification

A12 Model fit never shown; untargeted moments missed

> *"Both the model quantification and model validation are well below the bar for publication. The paper never shows the model fit of the quantitative model and the model does not match up to non-targeted moments, casting doubt on any counterfactual policy experiments."*

[Conceded.] Accepted without qualification. The calibration table now reports target, model value and gap for all 24 targeted moments. Two further moments are held out of the calibration entirely and reported as validation.

On the untargeted misses, the substantive ones that Referee 1 identifies in his comment 2c are now addressed at the source rather than defended. The homeless income ratio of 19.1 per cent from the linked administrative data is now a *target*, as are the income ratios for drug users and for people with a disorder. The excessive youth of the model's homeless population is addressed by the mental illness state, which generates persistent homelessness at older ages in a way that a pure income shock process could not.

A13 Key elasticities the model should match

> *"With respect to the policy evaluations, the key elasticities are (1) the elasticity of drug use to housing access (see R3) and (2) the elasticity of homelessness to drug addiction. The model should highlight these key elasticities in the analysis and show that the model is able to match them."*

This is the request that all four documents converge on, and it is the core of the new quantification.

The elasticity of drug use to housing access is now targeted directly, using Milby et al. (2005), which Referee 2 points to. The trial has two arms that differ in housing access for homeless adults with a substance use disorder, so the *contrast between arms* identifies the treatment effect and the *level in the no housing arm*, 4.48 consecutive weeks abstinent out of 24, or 18.7 per cent, identifies how costly abstention is for a household that already has a disorder. That second parameter was previously asserted to be small enough to keep the drug choice interior, which is a statement about numerical convenience rather than about data.

The elasticity of homelessness to substance use disorder is disciplined by the share of the homeless population with a drug use disorder, together with the employment penalty, which is now measured rather than asserted (see the letter to the Coordinating Editor, point 2 of section 3).

[Built since the first draft of this letter.] The housing access elasticity now has a parameter of its own, calibrated to the contrast between Milby's arms, and it is a *hazard* rather than a taste: recovery from a substance use disorder is six times harder on the street than in a home. I first proposed making the Stone-Geary shifter depend on the housing chosen and have retired that, because housing here is a choice, so a preference depending on it is a preference over a choice, and because the marginal value of the drug goes as $1/(b+v)$, so such a channel works by asserting that homelessness improves the experience of using. The hazard is the better home for it on the evidence as well as on the theory: a trial reports abstinence by arm, and abstinence by arm is a hazard. The consequence is that flow utility is now the same function of consumption, housing and drugs in every state and after every choice, which is what your A8 asked for. The two requests turn out to be one.

A14 Connect to Costantini and the other empirical work

> *"For example, Constantini (2025) finds for a sample of 300,000 Veterans that Housing First reduces three-year mortality by 4.6 percentage points while Treatment First has no long-term health effects, in contrast with the model simulations of this paper."*

[Point of correction.] I take this comment as the most serious in your letter, because if the reading is right the paper cannot survive it. Having worked through his design, I do not think it is right, and I want to set out why carefully rather than brush past it.

First, neither of his coefficients is my Experiment 1. Costantini estimates two local average treatment effects, both against *no housing intervention*. He has three treatment states, voucher, transitional housing, and neither, and two instruments: the leave out share receiving a voucher, and whether a Grant and Per Diem liaison performed the intake assessment in the week of arrival. Both of his estimates therefore live on the *generosity* margin, assistance against none. My Experiment 1 holds the number of assisted households fixed and varies only *conditionality*. His two coefficients do not identify the difference between two rationed regimes; that is why he compares them with a separate test rather than by subtracting one row from the other.

Second, his Treatment First is not my Treatment First. Grant and Per Diem is time limited transitional housing, a stay of six months to two years, after which the veteran must find their own housing. His own mechanism section attributes the null to that expiry. In his Figure 4, transitional housing recipients are *more* likely to be housed than voucher recipients in quarter one, because a bed is available immediately while a voucher has to find a landlord, and the effect then decays to 9 percentage points by quarter twelve while the voucher arm peaks at 34. The programme does not fail to house people; it stops housing them. My Treatment First is a permanent voucher whose *retention* is conditional on sobriety. These are different objects, and the difference is precisely the feature his mechanism section identifies as decisive.

[**New result.**] Rather than rest on that argument, I ask the model both of his questions, in his units, at his margin. Two arms, each with its own baseline and its own marginal expansion of access, each reporting the Wald ratio that an instrument estimates:

Arm	Model	Costantini
Permanent unconditional voucher	**[TO FILL: HF LATE]**	−4.60 pp, s.e. 1.7
Sobriety conditioned voucher expiring at the Grant and Per Diem rate	**[TO FILL: TF LATE]**	near zero

The expiry hazard is 0.8 per year, the reciprocal of the 1.25 year mean stay implied by his six month to two year window. Neither number is targeted, and I will report the result whichever way it comes out.

Third, and I did not expect this, his mechanism evidence supports the model's channel selection. His cause of death decomposition finds the effect running through exposure to the elements, assault, and blood borne infection from unsafe injection, and he states that Housing First has *no* effect on mental health care utilisation, ruling out mental health treatment as the channel. Those are exactly the channels this model has, and the one it deliberately does not. Independently, the At Home / Chez Soi trial finds housing moves mental illness symptom severity by nothing at any wave, and substance use problem severity by nothing either. Two very different designs agree.

On the other papers: Cohen (2024), Brounstein and Wieselthier (2026), Abramson (2026) and Humphries et al. (2026) are now discussed in the literature section, with Abramson (2026) and Humphries et al. (2026) positioned as the structural work on rental assistance that this paper's housing block is closest to.

On your final question, whether there are direct productivity effects of substance use disorder: yes, and it is now in the model. See A20.

A15 The fiscal cost of homelessness looks high

"The fiscal cost of homelessness for the United States is also estimated in recent work of Abramson and van Nieuwerburgh (forthcoming). It would be valuable to provide a comparison of their figures to the CEA estimates or provide robustness analysis with respect to these estimates of the fiscal costs. The current estimate appears high (see R3)."

[Point of correction.] I have read their paper and there is a point of fact worth stating, politely but clearly, because it affects what the comparison can do.

The 2,500 dollars per household per month for the literally homeless, and half that for the doubled up, is an *assumption* inside their calibration. They say so, and the components are listed in their footnote 11. It is not an independent estimate of the fiscal cost of homelessness, so it cannot serve as external validation of mine. Blended across their two groups it comes to about 1,775 dollars per month.

I report the comparison anyway, because a reader will want it, and I describe it accurately as a comparison of two calibration choices rather than of an estimate against a target. I also run the sensitivity, which both you and Referee 2 comment C6 ask for, and Referee 2's reading of the mechanism is correct: the cost of an unhoused household exceeds the rent on the smallest unit, which is what makes universal provision close to self financing. That is a result the paper should be transparent about rather than one it should bury, so the sensitivity is reported in the body.

[Built since the first draft of this letter.] **[Point of correction.]** You were right about the number, though not for the reason I first gave. My figure had been taken from Abramson and Van Nieuwerburgh's blended cost across the homeless and the precariously housed. The homeless state in my model is literal homelessness, shelters and the street, so the matching figure is their $2,500 per household per month rather than the $1,775 blend. The parameter has been changed.

A16 Where does 37 per cent of homeless with SUD come from?

"It is not clear where the statistic that 37 percent of homeless people are experiencing SUD comes from. This seems very high and will also depend greatly on the definition of homelessness . . . At a minimum, it should be clarified where this number is coming from and what definition of homelessness the author is targeting."

[Conceded.] You were right to be suspicious and the resolution is worse than a citation problem. The 37 per cent is the *alcohol* figure. In Gutwinski et al. (2021), which pools 39 studies of homeless people in high income countries, alcohol use disorder is 36.7 per cent. The drug use disorder figure from the same evidence base is 21.7 per cent. The paper is about drug use, so the target was simply the wrong row. It is now 21.7 per cent.

On the definition: the paper targets *literal* homelessness, sheltered and unsheltered, and excludes the doubled up. This is now stated explicitly, along with the reason, which is that the model's homeless state carries a mortality and employment penalty calibrated to cohorts of the literally homeless, so extending the definition without changing those penalties would be inconsistent.

While reworking this target, a second problem surfaced. The model's agent is a household, and the homelessness rate was built on a person count. Converting consistently gives 0.399 per cent rather than 0.30. Both corrections are documented in the letter to the Coordinating Editor.

Writing

A17 The calibration section reads as a long list

> *"Both the model parameterization and the calibration are rather thin. The section is monotonically written as a longlist of targets. . . . estimates appear to come from different decades, and it is often not clear which data waves are used (such as the SCF), making the paper impossible to replicate . . . there seems to be no distinction between parameters that are standard in the literature, versus those that are new to the paper. From the references, for example, it is not clear that the 10-year recovery rates for alcoholism (Vaillant, 1995) would apply to substance use disorder for the homeless."*

Rewritten. Three specific changes.

Every source now carries a vintage, and the parameters are split across two tables rather than one: the twenty nine that carry this paper's story and its policy analysis, and the twenty four that are standard in life cycle models. Each table marks every parameter as internally calibrated or externally set, and the text alongside gives, for each one, the direction in which the moment moves it and why that pairing is informative.

On replication specifically, since you say the paper is impossible to replicate as it stands: `data/DATA_APPENDIX.md` in the replication package now derives every number the calibration takes from outside the model, naming the dataset, its year, its universe and the computation that turns it into the figure the paper reports. It does the same for every validation moment, including the three places where the universe rather than the arithmetic decides the number. And the package carries the datasets themselves, the raw exports from HUD, CDC WONDER, the Federal Reserve, the Census, SSA, BLS and SAMHSA alongside the two microdata extracts, so nothing has to be downloaded to reproduce any of it.

The section is organised by *identification argument* rather than by parameter index: which moment moves which parameter, and why that pairing is informative. Referee 1 was complimentary about this argument in section 4.2 of the previous draft and asked for it to be shown numerically; that is comment 2f and it is done.

On Vaillant specifically: you are right that a ten year recovery rate for alcoholism in a general population is a poor guide to substance use disorder among the homeless, and it is no longer used that way. The housed recovery hazard and the homeless recovery multiplier are now identified jointly from Milby et al., whose sample is homeless adults with a substance use disorder, which is the population the parameter is about.

A18 Fewer counterfactuals in the body

> *"In terms of the policy analysis, ideally there would be fewer counterfactual analysis in the body of the paper to convey the main message. . . . The first and third experiment in this case seem to be most relevant singling out what is the better policy under rationed quantities and non-rationed settings."*

Done. Experiments 1 and 3 are in the body; the others are described in the text and reported in an appendix.

I would add one experiment to the body, and it is new. Referee 3's comment M2c asks where the market failure is if drug use is rational and forward looking. The answer is testable rather than

rhetorical, because the benchmark economy already insures homelessness through vouchers, food stamps and disability. Removing that insurance, and then removing it again with the drug choice frozen at its benchmark schedule, separates ordinary consumption smoothing moral hazard from moral hazard that runs through the *addiction* margin. I think this is the paper's most interesting single table and it directly answers the sharpest theoretical question in the reports.

A19 Language

> *"The language of 'addicted' seems incorrect. The author should modify throughout something to suffering from substance use disorder."*

Done throughout, in the text, the tables, the figures and the code.

A20 Direct productivity effects of substance use disorder

> *"Incidentally, are there direct productivity effects of substance use addiction on productivity?"*

[**New result.**] There are, and adding one solved a separate problem.

In the previous draft, substance use disorder affected employment *transitions* but not productivity conditional on working, so the entire income gradient among people with a disorder had to be produced by the employment margin. That is one channel carrying two jobs, and it is part of why the employment channel was calibrated too strong.

There is now a direct wage penalty. It is the parameter that carries the income ratio for people with a disorder, 68.7 per cent of average in NSDUH for ages 26 and above. Setting it to zero is reported as a robustness run rather than maintained as an assumption.

This change had a useful side effect. The value of life constant was previously calibrated to that same income ratio, which pins its *effect* on who uses drugs but says nothing about its *level*, and at the resulting parameters a majority of the population had negative continuation value. Moving the income ratio to the productivity penalty freed the constant to be pinned externally by the model implied value of a statistical life, which is what the paper always claimed to be doing.